

Part 2 of 3

READING THE DATASHEETS OF ELECTRONIC DEVICES

The device in focus this month is semiconductor diode. Depending upon the intended application, semiconductor diodes are broadly classified into general-purpose diodes/rectifiers, high-frequency/switching diodes and power diodes. Out of a wide range of performance specifications included in the datasheet of a typical semiconductor diode, some are more relevant for a particular application than the other

■ ANIL KUMAR MAINI AND
VARSHA AGRAWAL

The first part of this article highlighted the importance of reading a datasheet and the host of technical information contained therein. In particular, it focused on the salient features that are common to the datasheets of all categories of devices, illustrating the significance of different parameters with practical examples. In continuation, here we focus on semiconductor diodes.

Selection criteria for diodes

The specifications are the basis of selection criteria when it comes to choosing

the right device for a particular application. For instance, when choosing a diode for rectification purpose, the specifications to be taken into consideration are the peak inverse voltage (PIV), average forward current rating and the ampere-square seconds (I^2t) rating of the diode. Specifications such as reverse recovery time and switching speed are irrelevant.

Note that here we are referring to rectification of 50/60Hz AC mains, and the same is not true in the case of rectifiers used in switched-mode power supply applications. In fact, these parameters are not specified by the manufacturers in the datasheets of diodes intended for AC mains

rectification.

Similarly, for diodes used for switching applications, the important specifications to be considered are PIV rating, reverse recovery time and capacitance of the diode. Specifications like average rectified forward current do not have much significance.

Knowledge of these major specifications and their proper interpretation considerably helps a designer in choosing a diode for any given application. The specifications are defined in a datasheet under three major heads: absolute maximum ratings, electrical characteristics and thermal characteristics.

Absolute maximum ratings

Reverse breakdown voltage or the peak inverse voltage (V_{BR} , PIV). It is the maximum reverse voltage that a diode can withstand without breaking down. There are usually two different PIV ratings specified for diodes: repetitive peak inverse voltage (V_R or V_{RRM}) and non-repetitive peak inverse voltage (V_{RSM}). The non-repetitive rating is obviously greater than the repetitive rating. The one that needs to be considered depends upon the intended application. In a rectifier application, it is the repetitive inverse voltage rat-

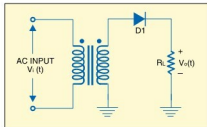


Fig. 1: Half-wave rectifier

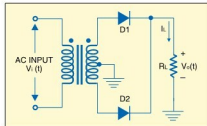


Fig. 2: Conventional full-wave rectifier

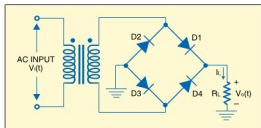


Fig. 3: Bridge rectifier

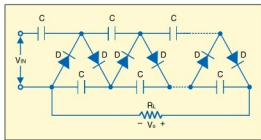


Fig. 4: Cascade voltage multiplier chain